Jiatao **Yan**

AI Postgraduate Student with a Focus on Computer Vision & Medical AI

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Summary

- Interests: multimodal clinical AI, interpretable modeling, and representation learning.
- Hands-on experience with deep learning pipelines (PyTorch/TensorFlow) for imaging and multi-modal data; familiar with reproducible development (Git, Docker, LaTeX).
- Kaggle Expert (Ranked 2,237th/204,979, 2 Bronze Medals); IEEE SMC 2025 Reviewer (CCF-C).

Education

Sussex Artificial Intelligence Institute, Zhejiang Gongshang University — *Hangzhou, China*

Master in Artificial Intelligence and Adaptive Systems (Distinction) — *Sept 2024 – Jan 2026*

- Overall average mark: 79.
- Thesis: Disentangling physics, anatomical, time, and identity information in latent variables of medical images.
- Focus: Alzheimer's disease; interpretable representation learning for medical images.
- Core Courses: Image Processing, NLP, ML, Intelligence in Animals and Machines, Python Programming.

Wenzhou Business College — *Wenzhou, China*

B.S. in Computer Science and Technology — *Sept 2019 – June 2023*

- GPA: 3.41/5.0 (84.7/100)
- Thesis: Smoking Behavior Detection Based on Deep Learning and Skeletal Frameworks.

Research & Internship Experience

National Innovation Training Program for University Students (Project No. 202213637004) — *Wenzhou, China*

Project Lead — *June 2022 – June 2023*

- First principal investigator of national-level research project. Responsible for technical design, team management, and project completion.

First Affiliated Hospital of Wenzhou Medical University - Hepatobiliary and Pancreatic Surgery Laboratory — *Wenzhou, China*

Research Intern — *Sept 2022 – Jan 2023*

- Developed medical image preprocessing software, obtained software copyright (2022SR0252378); filed invention patent (Appl. No. 202210991278.4).
- Contributed to deep learning model for leukemia diagnosis and exosome feature analysis in hepatocellular carcinoma research.

Zhejiang University Urban Planning and Design Institute — *Hangzhou, China*

Research Intern — *Sept 2025 – Apr 2026*

- Developed intelligent document processing and multi-agent workflow systems for government/enterprise scenarios.
- Introduced MCP tools and pluggable toolchains, implemented multi-agent collaboration and observability, added user interrupt and checkpoint resume.

Publications

Machine Learning Identifies Exosome Features Related to Hepatocellular Carcinoma

Published in *Frontiers in Cell and Developmental Biology* — *Sept 2022*

- Authors: Kai Zhu, Qiqi Tao, ***Jiatao Yan***, Zhichao Lang, Xinmiao Li, Yifei Li, Congcong Fan, Zhengping Yu
- DOI: 10.3389/fcell.2022.1020415
- Contribution (Co-first author, 3rd): Designed ML pipeline, applied Random Forest/SVM-RFE/LASSO algorithms to identify key biomarkers from exosome proteomics data.

Multi-omics and Machine Learning-driven CD8+ T Cell Heterogeneity Score for Prognosis

Published in *Molecular Therapy Nucleic Acids* — *Dec 2024*

- Authors: Di He, Zhan Yang, Tian Zhang, Yaxian Luo, Lianjie Peng, ***Jiatao Yan***, et al.
- DOI: 10.1016/j.omtn.2024.102413
- Contribution: Implemented LASSO regression and other ML algorithms to identify key genes for HNSCC prognosis, supporting CD8+ T cell heterogeneity score construction.

Using Multiomics and Machine Learning: Insights into Improving the Outcomes of Clear Cell Renal Cell Carcinoma via the SRD5A3-AS1/hsa-let-7e-5p/RRM2 Axis

PUBLISHED IN *ACS Omega* *June 2025*

- Authors: Mouyuan Sun, Zhan Yang, Yaxian Luo, Luying Qin, Lianjie Peng, Chaoran Pan, ***Jiatao Yan***, Tao Qiu, Yan Zhang
- DOI: 10.1021/acsomega.5c01337
- Contribution: Implemented ML algorithms to validate SRD5A3-AS1/hsa-let-7e-5p/RRM2 signaling axis role in clear cell renal cell carcinoma.

A multi-data fusion deep learning model for prognostic prediction in upper tract urothelial carcinoma

PUBLISHED IN *Frontiers in Oncology* *Aug 2025*

- Authors: Hongdi Sun, Siping Chen, Yongxing Bao, Fengyan You, Honghui Zhu, Xin Yao, Lianguo Chen, Jiangwei Miao, Fanggui Shao, Xiaomin Gao, Binwei Lin
- DOI: 10.3389/fonc.2025.1644250
- Contribution: Designed DL architecture for multi-phase CT analysis, developed methods to integrate imaging features with clinical data for survival prediction.

Manuscripts Under Review & In Preparation

Calibrated Guidance for Diffusion Policies in Offline Reinforcement Learning

Under Review, NeurIPS 2026 (CCF-A)

- Authors: Xiangyu Jiang, Chaofan Pan, ***Jiatao Yan***, Wei Wei, Xin Yang
- Proposed CGD to address distribution shift in offline diffusion policies via conformally-calibrated lower-confidence critic guidance; combines ensemble critic, split-conformal calibration, and state-adaptive scheduling to reduce OOD action rate by $3\times$ and outperform Diffusion-QL by +15.6 on OGBench trajectory-stitching.

Inspired by the Kidney Brain Co-Treatment Theory: Stagewise Lesion Aware Gated Fusion of Multimodal MRI for Acute Ischemic Stroke Lesion Segmentation

Under Review

- Contribution (Corresponding author): Proposed LAMF-UNet for acute ischemic stroke lesion segmentation (DWI/ADC/FLAIR), with staged fusion and lesion-prior guidance; achieved Dice 0.7942 on ISLES 2022, surpassing SegResNet/SwinUNETR baselines.

YOLOv11-LCDFS: Enhanced Smoking Detection With Low-light Enhancement

Under Review

- Authors: ***Jiatao Yan***, Zhuzikai Zheng, Zhengtan Yang, Hao Jiang, Peichen Wang, Fangjun Kuang, Siyang Zhang
- Contribution (First author): Developed YOLO-based architecture with low-light enhancement, loss function, attention mechanisms, and optimized upsampling.

A Spatio-Temporal Graph Transformer for Decoding Motor Imagery from fNIRS in Post-Stroke Patients

In Preparation

- Contribution (First author): Proposed STGT unifying channel topology and temporal dynamics for fNIRS motor imagery decoding; achieved 79.8% LOSO accuracy outperforming SVM/CNN/LSTM baselines; introduced attention-based interpretability to localize key channels and time windows as potential neural biomarkers.

A Deep Learning Model for Automated Sonographic Assessment of Diastasis Recti Abdominis

In Preparation

- Contribution (First author): Developed DL pipeline for automated ultrasound assessment of DRA, including rectus abdominis segmentation, keypoint estimation, and pixel-to-cm scale calibration for automatic inter-recti distance computation; supports near real-time inference.

Projects & Achievements

2026	**Bronze Medal, Ranked 97th/1424 (Top 7%)**, PhysioNet - Digitization of ECG Images	*Kaggle Competition*
2026	**Bronze Medal, Ranked 137th/1564 (Top 9%)**, Recod.ai/LUC - Scientific Image Forgery Detection	*Kaggle Competition*
2024	**Ranked 116th/3310 (Top 4%)**, Predict Podcast Listening Time	*Kaggle Competition*
2024	**Ranked 178th/4316 (Top 5%)**, Predict Calorie Expenditure	*Kaggle Competition*
May 2023	**Bronze Medal**, 18th ”Challenge Cup” Provincial College Student Competition	*Zhejiang Province*
Apr 2024	**Student Member Computer Innovation and Entrepreneurship Award**, Wenzhou Computer Society	*Wenzhou, China*

Patents & Intellectual Property

Software Copyright, Medical Image Computing Software	*2022SR0252378*
Software Copyright, Human Skeleton Recognition Software	*2022SR1258998*
Software Copyright, Cigarette Recognition Software	*2022SR1277520*
Software Copyright, Smoking Behavior Detection Software	*2022SR1277521*